CLOUD

CI/CD Pipelines Visual Guide

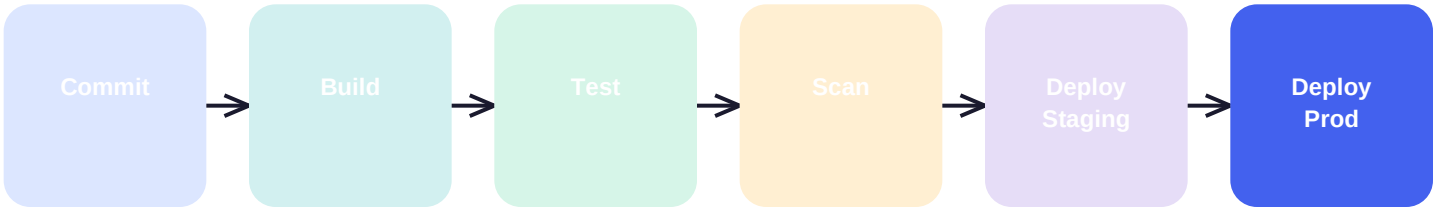
From git push to production
in minutes — safely, repeatedly, automatically.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Pipeline: Stages of Trust

Every commit climbs a ladder of verification.



Fail fast: cheap checks (lint, unit) run first; expensive ones (e2e, perf) run later. A red stage stops the line.

STEP 2/6

Continuous Integration

Merge small, merge often — let robots catch the breakage.

- Every push triggers: checkout → build → unit tests → lint/typecheck.
- Branch protection: no merge without green checks + review.
- Trunk-based development: short-lived branches (< 1 day) beat month-long feature branches.
- The build artifact (container image) is versioned once and promoted unchanged through stages.

Golden rule

If the build is red, fixing it is everyone's top priority.

STEP 3/6

Security Gates: Shift Left

Scan everything, early — before it reaches production.

SAST

Static analysis of source: injection, secrets, bad patterns

Dependency scan

Known CVEs in libraries (Dependabot, Snyk, Trivy)

Container scan

Image layers checked for vulns + misconfig

DAST / IAST

Running-app tests: fuzz inputs, probe auth

STEP 4/6

Deployment Strategies

How new versions reach users without drama.

Rolling

Replace instances gradually. Simple; mixed versions briefly.

Blue/Green

Two full environments; flip the router. Instant rollback.

Canary

1% → 10% → 50% → 100%, watching metrics at each step.

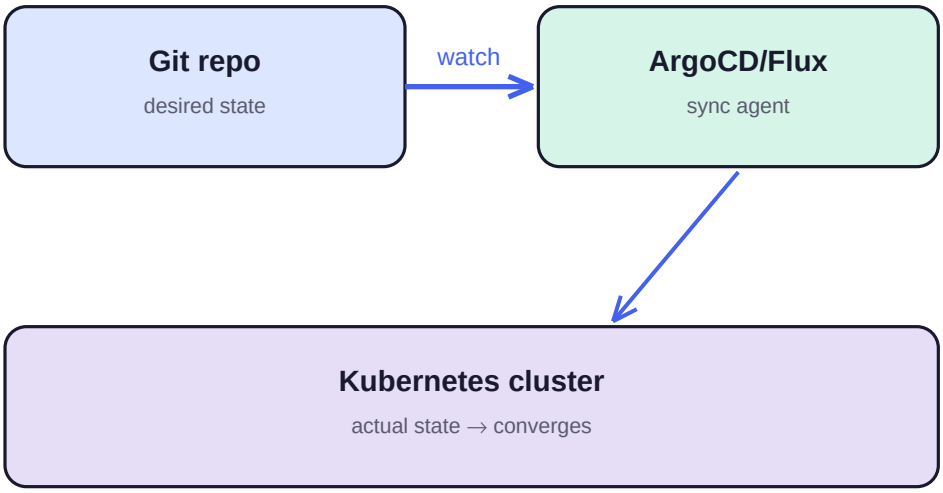
Feature flags

Deploy code dark, enable per-user/cohort at runtime.

STEP 5/6

GitOps: Git as the Source of Truth

The cluster converges to what's in git — automatically.



Rollbacks = git revert. Drift (manual kubectl edits) gets auto-corrected.

Pull requests become the change-management audit trail.

STEP 6/6

DORA Metrics: Are We Actually Good at This?

Four numbers that predict engineering performance.

Deploy frequency

How often to prod — elite: on demand

Lead time

Commit → prod — elite: < 1 hour

Change fail rate

% of deploys causing incidents — elite: < 15%

MTTR

Restore service after incident — elite: < 1 hour

STEP ✓I✓

Key Takeaways

CI/CD — Remember

- Pipeline stages climb in cost: lint/unit first, e2e/perf later; red stops the line.
- Build the artifact once; promote the identical image through every stage.
- Shift security left: SAST, dependency, container, and DAST scans in-pipeline.
- Deploy progressively: rolling, blue/green, canary, feature flags.
- GitOps makes git the source of truth; DORA metrics tell you if it's working.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev